

Notes: Naritomi (AER, 2019)

Consumers as Tax Auditors

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Institutional background	3
3.1	Sao Paulo and the VAT	3
3.2	Nota Fiscal Paulista	3
4	Data	4
4.1	Firm data	4
4.2	Receipt, complaint, and consumer data	4
5	Conceptual framework	4
5.1	Government monitoring only	4
5.2	Adding consumer monitoring	5
5.3	Comparative statics	5
6	Empirical design	6
6.1	Retail versus wholesale difference-in-differences	6
6.2	Firm-level DD	6
6.3	Heterogeneous effects	7
6.4	Complaint event study	7
6.5	Lottery event study	7
7	Tables and figures	7
7.1	Table 1 and Figure 1: Data and NFP take-up	7
7.2	Figure 2 and Table 2: Main reported-revenue effects and mechanisms	8
7.3	Figure 3 and Figure 4: Whistle-blower effects and reward salience	9
7.4	Figure 5 and Figure 6: Lottery behavior and tax revenue	10
7.5	Table 3: Tax liabilities, reported expenses, and value added	10
8	Short summary	11

9	Conclusion	12
9.1	Core conclusion	12
9.2	Economic intuition	12
9.3	Incremental contribution revisited	12
9.4	Policy implications	12
9.5	Limitations	12
9.6	Takeaway	12

1 Big picture

- **Question.** Can governments improve tax compliance in hard-to-tax retail markets by turning consumers into third-party monitors of firm sales?
- **Policy setting.** The paper studies *Nota Fiscal Paulista* (NFP), a program introduced in Sao Paulo, Brazil, in October 2007. The program rewards consumers for asking firms to issue receipts that include the consumer's tax identification number.
- **Core enforcement problem.** The value-added tax (VAT) has a self-enforcing logic in business-to-business transactions because buyers want receipts to claim input credits. This mechanism breaks down at the final-consumer stage because consumers usually have no direct monetary reason to request receipts.
- **Policy innovation.** NFP gives final consumers tax rebates and lottery tickets tied to receipt requests. It also creates an online system where consumers can verify whether firms reported their receipts and can file complaints.
- **Main empirical strategy.** The paper compares retail sectors, which sell mostly to final consumers and are highly exposed to NFP, with wholesale sectors, which sell mostly to other firms and are less exposed. This is implemented through difference-in-differences and event-study designs.
- **Main result.** Reported revenue in retail rises by at least 21 percent over four years relative to wholesale. Firm-level estimates are similar and imply a large increase in reported sales compliance.
- **Mechanism 1: whistle-blower threats.** Firms respond more when they face many distinct consumers. Firms also report more receipts and more revenue after receiving their first complaint from a consumer.
- **Mechanism 2: collusion costs.** The effect is larger in sectors with many transactions and small-ticket purchases. This is consistent with fixed costs of concealing collusion with many consumers.
- **Mechanism 3: behavioral response to lotteries.** Consumers pay attention to lottery draws and tax-rebate disbursements. Even small lottery wins increase future receipt requests and the total value of receipts.
- **Fiscal result.** Firms increase reported expenses, but not enough to offset the increase in reported revenue. Tax liabilities rise, and net tax revenue rises by about 9.3 percent after accounting for consumer rewards.
- **Economic takeaway.** Third-party information can be created, not merely passively observed. Governments can use consumers as decentralized tax auditors when they design incentives and whistle-blowing channels that make collusion difficult.

2 Incremental contribution of the paper

- **From third-party reporting to third-party creation.** Existing work shows that third-party information is crucial for compliance. This paper studies a policy that actively creates third-party information trails by giving consumers incentives to ask for receipts.

- **The last-mile VAT problem.** The paper directly studies the point where VAT self-enforcement is weakest: the final transaction between a firm and a consumer.
- **Collusion as the central friction.** The paper does not assume that third-party information automatically works. It asks why consumers and firms may fail to collude, emphasizing whistleblower risk and fixed costs of concealing collusive discounts.
- **Unique administrative data.** The paper uses de-identified administrative tax records, receipt-level data, consumer participation records, complaint records, and lottery outcomes from Sao Paulo’s tax authority.
- **Multiple identification layers.** The paper combines sector-level DD, firm-level DD, a complaint event study, lottery-event comparisons, and fiscal accounting.
- **Consumer-side evidence.** Unlike many tax enforcement papers focused only on firms, this paper directly studies how consumers respond to incentives and how their response changes firm reporting.
- **Behavioral public finance angle.** The lottery evidence suggests that reward salience and probability misperception may amplify enforcement incentives at low fiscal cost.
- **Policy contribution.** The paper provides evidence that receipt rewards can raise net tax revenue even after paying rewards, and it clarifies which design features matter: verification, rewards, complaint channels, and salience.

3 Institutional background

3.1 Sao Paulo and the VAT

Sao Paulo is Brazil’s largest state and accounts for a large share of national GDP. The main state tax is the ICMS, a value-added tax on goods and some services. The standard ICMS rate is 18 percent, and it is administered by the state tax authority. Formal firms submit tax returns monthly.

The enforcement challenge is especially severe in retail. Business-to-business transactions generate paper trails because buyers have incentives to request invoices for input credits. Final consumers usually have no such incentive. As a result, retailers can underreport final sales unless the state has direct audit evidence or receipt information.

3.2 Nota Fiscal Paulista

NFP was introduced in October 2007 to reduce evasion and promote fiscal citizenship. The program has four main components:

- **Receipt identification.** Consumers can request that their SSN-equivalent taxpayer ID be printed on the receipt.
- **Tax rebates.** Consumers receive a share of the VAT paid by the retailer. The average rebate is around 1 percent of the purchase value.
- **Lotteries.** Consumers receive lottery tickets based on receipt spending; every US\$50 of eligible purchases generates one lottery ticket.

- **Online verification and complaints.** Consumers can log into an online account, verify whether receipts were correctly reported, and file complaints against firms.

The program was phased in from October 2007 to May 2008. The complaint system became available in October 2008, monthly lottery draws began in December 2008, and the first tax rebates were disbursed in April 2009.

4 Data

4.1 Firm data

The firm data come from administrative tax records for Sao Paulo from January 2004 to December 2011. The main sample includes firms registered in retail and wholesale sectors. Outcomes are aggregated by seven-digit sector for the sector-level DD analysis and also used at the firm level for firm fixed-effect regressions.

Key variables include:

- monthly reported gross revenue;
- tax liabilities, for sectors where measured tax due best approximates actual liability;
- reported inputs and value added for firms continuously in the VAT system;
- firm characteristics such as sector, size, and legal structure.

4.2 Receipt, complaint, and consumer data

Receipt microdata are available from January 2009 to December 2011. They identify receipts with consumer SSNs, total receipt value, establishment identifiers, and timing. These data allow the paper to measure consumer take-up and firm reporting with SSN receipts.

Complaint data come from the online NFP system. The paper observes when consumers file complaints against establishments and uses the first complaint as an event to examine firm reporting responses.

Consumer data include participation in lotteries, number of lottery tickets, lottery prize values, number of receipts requested, and total value of SSN receipts. The lottery analysis focuses on consumers who participated in lotteries between July 2009 and June 2011.

5 Conceptual framework

5.1 Government monitoring only

A firm has true revenue $\bar{Y}$ and reports revenue Y . Evasion is

$$E = \bar{Y} - Y.$$

The firm pays tax rate τ on reported revenue and faces a probability of detection

$$p = a(E)d,$$

where $a(E)$ is the audit probability and d is the government's ability to detect evasion conditional on audit. If caught, the firm pays a fine proportional to evaded taxes.

The firm's expected profit under government monitoring only can be written as

$$\pi = (\bar{Y} - \tau Y)(1 - p) + [(1 - \tau)\bar{Y} - \theta\tau(\bar{Y} - Y)]p.$$

The interior first-order condition is

$$[a(E) + a'(E)E]d(1 + \theta) = 1.$$

Interpretation. The marginal cost of evasion rises with the probability that the government detects evasion. If third-party information improves detection ability d , firms report more.

5.2 Adding consumer monitoring

NFP gives consumers rewards for receipts and lets them act as whistle-blowers. If firms want to keep evading, they must either refuse receipts or offer discounts to induce consumers not to request receipts. This creates two problems for firms:

- firms must share some evasion rents with consumers;
- firms reveal evasion information to many consumers, any of whom may blow the whistle.

Let $\kappa(\alpha)$ denote the value consumers place on rewards, where α is the reward rate. Let N be the number of consumers. If each consumer may blow the whistle with probability ε , consumer monitoring raises the detection ability to

$$d_c(N) = 1 - (1 - d)(1 - \varepsilon)^N.$$

The firm's expected profit becomes

$$\pi = (\bar{Y} - \tau Y)(1 - p_c) + [(1 - \tau)\bar{Y} - \theta\tau(\bar{Y} - Y)]p_c - \kappa(\alpha)\tau(\bar{Y} - Y) - \rho N.$$

The interior first-order condition is

$$[a(E) + a'(E)E]d_c(1 + \theta) = 1 - \kappa(\alpha).$$

Interpretation. Consumer rewards lower the benefit of evasion because the firm must compensate consumers. Consumer whistle-blowing raises the cost of evasion because more informed parties increase detection risk.

5.3 Comparative statics

The framework generates several empirical predictions:

- **Higher consumer reward valuation** should reduce evasion because firms must match more valuable rewards to collude.
- **More distinct consumers** should increase compliance because more consumers raise the whistle-blower threat.
- **More transactions** should increase compliance because fixed costs of concealment are paid repeatedly.
- **Lower transaction value** should increase the policy effect for a given firm size because the firm must conceal more transactions.
- **Whistle-blower events** should increase compliance if firms update their beliefs about consumer complaints and detection probability.
- **Smaller firms** should respond more because they likely have lower baseline third-party information trails and lower baseline detection risk.

6 Empirical design

6.1 Retail versus wholesale difference-in-differences

The main sector-level specification is

$$\log R_{st} = \eta_s + \gamma_t + \sum_{k=-8}^8 \beta_k (Treat_s \times Period_t^k) + u_{st},$$

where R_{st} is reported revenue in sector s and month t , $Treat_s$ indicates retail sectors, and $Period_t^k$ are six-month windows around the implementation of NFP.

The average DD version is

$$\log R_{st} = \eta_s + \gamma_t + \beta (Treat_s \times Post_t) + u_{st}.$$

Identification. The key assumption is that, absent NFP, reported revenue in retail and wholesale sectors would have followed parallel trends. The long pre-period allows the paper to show that the two groups had similar trends before the policy.

6.2 Firm-level DD

The firm-level specification collapses data into pre- and post-periods:

$$\log R_{its} = \eta_i + \gamma Post_t + \beta (Treat_s \times Post_t) + \varepsilon_{its}.$$

Firm fixed effects absorb time-invariant firm heterogeneity. The regressions are weighted by pre-policy reported revenue, so large economic sectors/firms receive greater weight in approximating aggregate effects.

6.3 Heterogeneous effects

To test mechanisms, the paper interacts the DD treatment with firm or sector characteristics:

$$\log R_{its} = \eta_i + \gamma Post_t + \sum_{m=1}^2 \alpha_m (d_{ms} \times DD_{ts}) + f(x_i) DD_{ts} + \varepsilon_{its}.$$

The heterogeneity variables include firm size, volume of different consumers, volume of transactions, and value of transactions.

6.4 Complaint event study

The whistle-blower analysis compares firms that receive their first complaint to firms that have not yet received a complaint at the same event date. The specification is

$$\log Y_{iekt} = \gamma_{ie} + \pi_t + \phi_k + \beta I\{k \geq 0, ie = T\} + u_{iekt},$$

where k is event time relative to the first complaint.

6.5 Lottery event study

The lottery analysis compares winners of small prizes to non-winners with the same number of lottery tickets. The specification is

$$y_{je kt} = \gamma_{je} + \pi_t + \phi_k + \beta I\{k \geq 0, je = Win\} + u_{je kt}.$$

The key idea is that winning a small prize should not create a large income effect, so a persistent increase in receipt requests is interpreted as a behavioral response to lottery salience or perceived winning probability.

7 Tables and figures

7.1 Table 1 and Figure 1: Data and NFP take-up

Read:

- Table 1 summarizes the firm and consumer samples. Retail firms have far more SSN receipts and distinct consumers than wholesale firms, consistent with the treatment-intensity logic.
- The consumer sample shows active participation: consumers ask for receipts, claim tax rebates, receive lottery tickets, and sometimes win lottery prizes.
- Figure 1 shows that the number of SSN receipts is much larger in retail than wholesale.
- Figure 1 also shows that by 2011, SSN receipts cover roughly 40 percent of retail reported revenue, but less than 9 percent of wholesale revenue.

Economic message: Retail is much more exposed to consumer monitoring than wholesale. This validates the retail-versus-wholesale treatment-intensity design.

TABLE 1—DESCRIPTIVE STATISTICS

	Observations	Mean	SD	Time period
<i>Firm sample</i>				
Retail firms				
Reported revenue	37,717,931	18,209.68	459,832	Jan. 2004–Dec. 2011
Number of receipts with SSN	12,838,202	174.30	12,925	Jan. 2009–Dec. 2011
Number of consumers	12,838,202	106.18	7,047	Jan. 2009–Dec. 2011
Wholesale firms				
Reported revenue	3,494,539	95,147.40	384,970	Jan. 2004–Dec. 2011
<i>Consumer sample</i>				
Number of receipts	181,032,084	8.75	9.23	Jan. 2009–Dec. 2011
Number of businesses	181,032,084	5.24	5.16	Jan. 2009–Dec. 2011
Tax rebate	181,032,084	4.66	8.18	Jan. 2009–Dec. 2011
Total expenditure in SSN receipts	181,032,084	502.89	1,060.49	Jan. 2009–Dec. 2011
Number of lottery tickets	181,032,084	1.40	4.86	Jan. 2009–Dec. 2011
Lottery prize value	181,032,084	0.57	3.33	Jan. 2009–Dec. 2011

Notes: Table 1 presents descriptive statistics of the main variables for each data source: firms and consumers. All values are in US dollars (US\$1 = RS2) and are measured monthly. Reported revenue is the monthly gross reported revenue by firms. Number of receipts with SSN is the monthly number of receipts firms report to the tax authority with the Social Security Numbers (SSNs) of the consumer. Number of consumers is the total number of different consumers who have been in receipt of a receipt. The consumer sample includes consumers that are in receipt of a receipt.

(a) Table 1: Descriptive statistics.

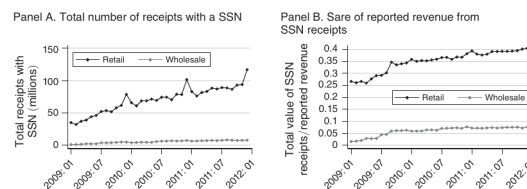


FIGURE 1. NFP RECEIPTS AND REVENUE SHARES: RETAIL VERSUS WHOLESALE

Notes: Panel A shows the aggregate number of SSN receipts issued by firms in retail versus wholesale sectors. The figure plots the raw data. The spikes around December of each year follows the seasonal variation in consumption. Panel B shows the share of the total reported revenue in retail that is covered by SSN receipts in retail sectors and wholesale sectors. Even though the program was created in October 2007, the data from the NFP program available for the analysis begins in January 2009. For more details, see Section II and online Appendix B. The two figures

(b) Figure 1: NFP receipts and revenue shares.

7.2 Figure 2 and Table 2: Main reported-revenue effects and mechanisms

Read:

- Figure 2 shows that retail and wholesale reported revenue move similarly before NFP. After the policy begins, retail reported revenue rises relative to wholesale.
- The event-study coefficients in Figure 2 are flat before the policy and rise gradually afterward, consistent with gradual consumer take-up.
- The DD estimate in Figure 2 is 0.208, implying roughly a 21 percent increase in reported revenue.
- Table 2 shows a firm-level DD estimate of 0.254, or about a 25 percent increase in reported revenue.
- Table 2 also shows that the effect is stronger for small firms, high-consumer-volume sectors, high-transaction-volume sectors, and low-ticket-value sectors.

Economic message: NFP substantially increases reported revenue, and the heterogeneity pattern supports the mechanism that consumer monitoring works through whistle-blower threats and collusion costs.

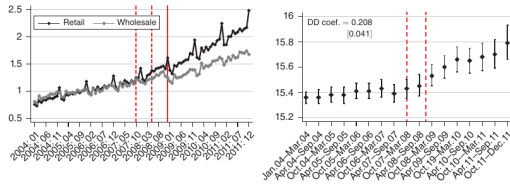


FIGURE 2. EFFECT OF THE POLICY ON REPORTED REVENUE: RETAIL VERSUS WHOLESALE

Notes: Panel A shows reported revenue changes for retail and wholesale sectors. Each line is the revenue reported by all firms aggregated by retail or wholesale scaled by the average monthly reported revenue before October 2007 for each sector group. The figure plots the raw data. The spikes around December of each year follows the seasonal variation in consumption. The vertical lines highlight the key dates for the implementation of the NFP program.

(a) Figure 2: Effect of the policy on reported revenue.

	log reported revenue				
	(1)	(2)	(3)	(4)	(5)
DD (Post October 07 $\times$ retail)	0.254 [0.0722]				
DD $\times$ large firms		0.253 [0.0732]			
DD $\times$ small firms		0.350 [0.0511]			
DD $\times$ high volume of different consumers			0.246 [0.0705]		
DD $\times$ low volume of different consumers			0.0329 [0.0919]		
DD $\times$ high volume of transactions				0.253 [0.0335]	
DD $\times$ low volume of transactions				0.018 [0.0391]	
DD $\times$ high value of transactions					0.097 [0.0689]
DD $\times$ low value of transactions					0.285 [0.0754]
3rd-order polynomial of firm size $\times$ DD			X	X	X
Time fixed effects	X	X	X	X	X
Firm fixed effects	X	X	X	X	X
Observations	1,035,268	1,035,268	1,035,268	1,035,268	1,035,268
Adjusted R^2	0.907	0.907	0.908	0.909	0.908

Notes: Table 2 displays the main coefficients from firm regressions. The variable DD is defined as the interaction

(b) Table 2: Reported revenue effects and heterogeneity.

7.3 Figure 3 and Figure 4: Whistle-blower effects and reward salience

Read:

- Figure 3 studies firm behavior after the first complaint. Firms receiving a complaint increase the number of receipts by about 7 percent and reported revenue by about 3 percent relative to matched no-complaint firms.
- This suggests that complaints reveal that consumers can act as monitors and cause firms to increase compliance.
- Figure 4 shows that consumers pay attention to program incentives. Google searches for NFP spike around lottery-result dates.
- Figure 4 also shows spikes in reward claims when tax rebates are disbursed, especially in April and October.

Economic message: The policy works on both sides of the market: firms respond to consumer complaints, and consumers respond to reward schedules.

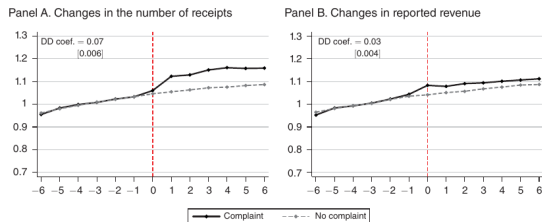


FIGURE 3. WHISTLE-BLOWER EFFECT ON FIRM COMPLIANCE

Notes: Panels A and B plot the changes in the total number of receipts firms issue and changes in reported revenue to the government before and after a firm receives the first complaint. Both graphs display changes across

(a) Figure 3: Whistle-blower effect on firm compliance.

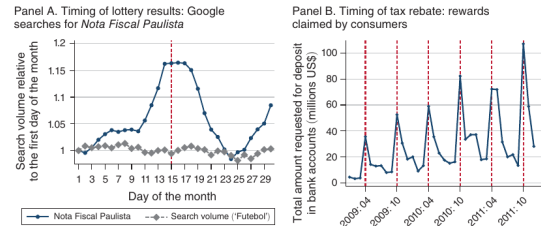


FIGURE 4. ARE CONSUMERS PAYING ATTENTION TO THE REWARDS SCHEDULE?

Notes: Panel A displays the search volume from Google Trends website for Google searches with terms related to

(b) Figure 4: Consumer attention to reward schedules.

7.4 Figure 5 and Figure 6: Lottery behavior and tax revenue

Read:

- Figure 5 compares winners and non-winners of small lottery prizes with similar winning odds.
- A US\$5 lottery win increases the future number of receipts by about 0.0725 and the total value of receipts by about US\$2.507 per month.
- The effect persists for at least six months, and the cumulative increase in receipt value exceeds the prize amount.
- Figure 6 shows that tax liabilities rise in retail relative to wholesale after NFP, with a DD estimate of 0.259.
- Figure 6 also shows that Sao Paulo's VAT revenue as a share of GDP increases slightly relative to other Brazilian states after the program.

Economic message: Lotteries may amplify consumer participation beyond their expected monetary value. The program raises reported tax liabilities, not just reported sales.

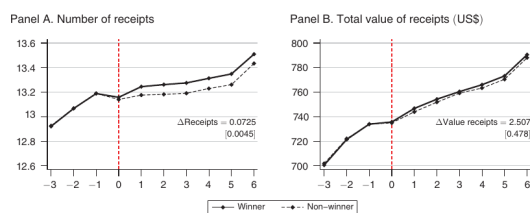


FIGURE 5. THE EFFECT OF A 5-DOLLAR LOTTERY WIN ON CONSUMER PARTICIPATION

Notes: The graphs show the raw data by month aggregating all lotteries from June 2009 to June 2011. The x-axis is the number of months since the individual participated in a lottery. The winner group got a cash prize of US\$5 (R\$10) and the non-winner group did not get any prize. Panel A plots the total number of receipts consumers

(a) Figure 5: Effect of a US\$5 lottery win on consumer participation.

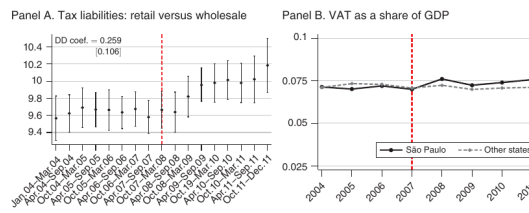


FIGURE 6. EFFECT OF THE POLICY ON TAX REVENUE

Notes: Panel A plots regression coefficients from estimating specification (5) using log of tax liabilities as the dependent and a sample of sectors for which total tax due best approximates the tax liability of firms between January 2004 and December 2011 (see online Appendix B for more detail). Similarly, the difference-in-differences

(b) Figure 6: Effect of the policy on tax revenue.

7.5 Table 3: Tax liabilities, reported expenses, and value added

Read:

- Panel A shows that reported tax liabilities increase by about 31.6 percent in the firm-level tax sample.
- The extensive margin of positive tax liabilities rises by 0.043 but is not statistically significant.
- Panel B studies firms continuously in the VAT system. Reported revenue rises by 0.363, reported inputs rise by 0.302, and value added rises by 0.387.
- This means firms do adjust reported inputs, but not enough to undo the revenue effect.
- Combining the tax liability increase with reward costs, the paper calculates that tax revenue rises by 9.3 percent net of rewards, and by 17.6 percent if unclaimed rewards are taken into account.

Economic message: Enforcement of reported sales can increase tax revenue even when firms adjust reported expenses. The VAT credit-chain limits complete offsetting through input claims.

TABLE 3—TAX LIABILITY AND REPORTED EXPENSES: RETAIL VERSUS WHOLESALE

	log of reported revenue (1)	log of tax liability (2)	Positive tax liability (3)	
<i>Panel A. Tax sample</i>				
DD (post October 07 $\times$ retail)	0.311 [0.151]	0.316 [0.137]	0.043 [0.0350]	
Firm fixed effects	X	X	X	
Time fixed effects	X	X	X	
Observations	167,110	133,950	167,110	
Adjusted R^2	0.85	0.876	0.801	
	log of reported revenue (1)	log of reported inputs (2)	log of reported value added (3)	Positive value added (4)
<i>Panel B. Expenses, output and value added—firms that were always VAT</i>				
DD (post October 07 $\times$ retail)	0.363 [0.0824]	0.302 [0.0833]	0.387 [0.105]	0.019 [0.0153]
Firm fixed effects	X	X	X	X
Time fixed effects	X	X	X	X
Observations	88,422	88,422	70,845	88,422
Adjusted R^2	0.87	0.85	0.90	0.71

Notes: Table 3 displays the main coefficients from regressions described in Section V using the firm-level data. The variable *DD* is defined as the interaction between a dummy for retail sectors (*Retail*) and a dummy that equals 1 for time periods after October 2007 (*Post October 07*). The data are collapsed into two periods: before and after October 2007. Time and firm fixed effects are included in all regressions. The regressions are dollar-weighted (each observation is weighted by the pre-policy reported revenue) such that each observation contributes to all regres-

Table 3: Tax liability and reported expenses.

8 Short summary

- The paper studies Nota Fiscal Paulista, a Sao Paulo program that rewards consumers for asking firms to report final sales through receipts.
- The policy addresses the last-mile problem of the VAT, where final consumers normally have no incentive to request receipts.
- The main empirical strategy compares retail sectors to wholesale sectors before and after program implementation.
- Retail reported revenue increases by at least 21 percent over four years.
- Firm-level DD estimates are similar and show about a 25 percent increase in reported revenue.
- The effect is stronger for smaller firms, sectors with more distinct consumers, sectors with more transactions, and sectors with smaller ticket values.
- Firms increase reporting after receiving the first consumer complaint: receipts rise by about 7 percent and revenue by about 3 percent.
- Consumers respond to incentives: Google searches and reward claims spike around reward dates.
- Small lottery wins cause consumers to request more receipts and spend more through SSN receipts for several months.
- Firms increase reported inputs, but reported value added and tax liabilities still increase.
- Net tax revenue rises by about 9.3 percent after accounting for rewards.

9 Conclusion

9.1 Core conclusion

Naritomi shows that governments can improve tax enforcement by creating consumer-side incentives for third-party reporting. In Sao Paulo, rewarding consumers for receipts raises firm-reported sales, increases tax liabilities, and generates positive net revenue despite the cost of rewards.

9.2 Economic intuition

The policy changes the game between firms and consumers. Before NFP, final consumers had little reason to ask for receipts, so firms could underreport sales. After NFP, consumers have direct incentives to request receipts and tools to verify reporting. Firms can try to collude with consumers, but collusion requires discounts, repeated concealment, and exposure to whistle-blowers. These frictions make compliance more attractive.

9.3 Incremental contribution revisited

The paper contributes by showing that third-party information trails can be endogenously produced through policy design. It also shows that the enforcement value of third-party information depends on whether collusion is difficult. Finally, it adds a behavioral public finance mechanism: lottery rewards can raise the perceived value of participation and increase monitoring effort at relatively low monetary cost.

9.4 Policy implications

The results suggest that consumer reward programs can be useful in retail sectors where evasion is difficult to detect through normal audits. The most effective designs likely combine small monetary rewards, salient lotteries, low participation costs, online verification, and credible complaint channels. The findings are especially relevant for VAT systems in developing economies, where retail-stage evasion can weaken the whole invoice-credit chain.

9.5 Limitations

The DD design relies on retail and wholesale following parallel reporting trends absent the policy. The evidence supports this assumption visually and through robustness checks, but the design is still not randomized. The welfare analysis is also incomplete because the data do not fully measure administrative costs, consumer time costs, psychological costs or benefits of lottery participation, and long-run changes in tax morale.

9.6 Takeaway

The enduring takeaway is that consumers can become effective tax auditors when the government gives them incentives, information, and a whistle-blowing channel. In this setting, decentralized

consumer monitoring raises compliance and net revenue by making hidden retail evasion harder to sustain.